

MEMO: DESCRIBING PERFECT MAPS WITH PULLBACKS

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In this note, following [1], we describe a characterization of perfect maps by means of pullbacks. For the parts I did not understand very well, I also referred to [3].

The English in this manuscript was written with the assistance of AI tools, including Codex.

1. PULLBACKS

I assume that the readers already know commutative diagrams.

Let X, Y, Z be topological spaces, and consider maps $f: X \rightarrow Z$ and $g: Y \rightarrow Z$.

$$\begin{array}{ccc} & X & \\ & \downarrow f & \\ Y & \xrightarrow{g} & Z \end{array}$$

In this situation, a topological space P together with two maps $g': P \rightarrow X$ and $f': P \rightarrow Y$ is called a **pullback** of the above diagram if the following diagram is commutative,

$$\begin{array}{ccc} P & \xrightarrow{g'} & X \\ \downarrow f' & & \downarrow f \\ Y & \xrightarrow{g} & Z \end{array}$$

and, whenever another T , $b: T \rightarrow X$, and $a: T \rightarrow Y$ making the diagram commute are given, there exists a unique u making the following diagram commute.

$$\begin{array}{ccccc} T & & & & \\ & \searrow u & \xrightarrow{b} & & \\ & & P & \xrightarrow{\quad} & X \\ & \searrow a & \downarrow & & \downarrow f \\ & & Y & \xrightarrow{g} & Z \end{array}$$

To indicate that P is a pullback, we write a right-angle symbol as follows.

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$$\begin{array}{ccc}
P & \xrightarrow{g'} & X \\
\downarrow f'^\perp & & \downarrow f \\
Y & \xrightarrow{g} & Z
\end{array}$$

The map f' is sometimes called a base change. The pullback P is sometimes denoted by $X \times_{Z, f, g} Y$ or $X \times_Z Y$.

From the above definition, we see that a pullback, if it exists, is unique up to isomorphism. We now check several basic facts about pullbacks of topological spaces.

Theorem 1. *In the category of topological spaces, pullbacks always exist. In other words, in the situation above, the pullback P always exists. Moreover, as a concrete description of the pullback, we may take $P = \{ (x, y) \in X \times Y \mid f(x) = g(y) \}$, with f' and g' the projections.*

Proof. Since we have given a concrete description, it is enough to check commutativity and the universal property. If you move your hands and compute briskly, it becomes clear. $\square$

Although pullbacks of topological spaces can be written concretely as above, what seems important is to know what they can be identified with in special situations.

Proposition 2. *The following statements hold.*

- (A1) *In the following diagram, $f^{-1}(x)$ is a pullback. That is, for a map $f: X \rightarrow Z$, when one considers a point $y: \{*\} \rightarrow Z$, the base change of f is the generalized fiber $f^{-1}(y)$.*

$$\begin{array}{ccc}
f^{-1}(y) & \longrightarrow & X \\
\downarrow & \lrcorner & \downarrow f \\
\{*\} & \xrightarrow{y} & Y
\end{array}$$

- (A2) *For a continuous map $f: X \rightarrow Y$ and a topological space S , the following diagram is a pullback.*

$$\begin{array}{ccc}
X \times S & \longrightarrow & X \\
\downarrow f \times 1_S & & \downarrow f \\
Y \times S & \xrightarrow{pr} & Y
\end{array}$$

For a continuous map $f: X \rightarrow Y$ and a subset B of Y , the following diagram is a pullback. Here $\iota: B \rightarrow Y$ is the inclusion map.

(A3)

$$\begin{array}{ccc} f^{-1}(B) & \longrightarrow & X \\ \downarrow & \lrcorner & \downarrow f \\ B & \xrightarrow{\iota} & Y \end{array}$$

(A4) *For a continuous map $f: X \rightarrow Y$, the following trivial pullback diagram holds.*

$$\begin{array}{ccc} X & \xrightarrow{1_X} & X \\ \downarrow f & \lrcorner & \downarrow f \\ Y & \xrightarrow{1_Y} & Y \end{array}$$

(A5) *Suppose that the following diagram is a pullback.*

$$\begin{array}{ccc} P & \xrightarrow{h} & X \\ \downarrow v & \lrcorner & \downarrow f \\ Y & \xrightarrow{g} & Z \end{array}$$

Then for any topological space S , the following diagram is also a pullback.

$$\begin{array}{ccc} P \times S & \xrightarrow{h \times 1_S} & X \times S \\ \downarrow v \times 1_S & \lrcorner & \downarrow f \times 1_S \\ Y \times S & \xrightarrow{g \times 1_S} & Z \times S \end{array}$$

Proof. Proof of (A1): In the above situation, $X \times_Z \{*\} = \{(x, *) \in X \times \{*\} \mid f(x) = y\}$, so $X \times_Z \{*\}$ and $f^{-1}(y)$ are homeomorphic.

Proof of (A2): This follows by briskly computing the universal property.

Proof of (A3): We have $P = X \times_Y B = \{(x, b) \in X \times B \mid f(x) = \iota(b)\}$. Define $\alpha: f^{-1}(B) \rightarrow P$ by $\alpha(x) = (x, f(x))$. Define $\beta: P \rightarrow f^{-1}(B)$ by $\beta(x, b) = x$. These maps are inverse to each other, so P and $f^{-1}(B)$ are homeomorphic.

Proof of (A4): This follows by computing the universal property.

Proof of (A5): This follows by briskly computing the universal property.

In general, pullbacks commute with limits. In particular, pullbacks commute with products, so the product of two pullback diagrams is again a pullback.

Let us end the proof around here. □

Connecting pullbacks with each other again gives a pullback.

Theorem 3. *Consider the situation of the following commutative diagram.*

$$\begin{array}{ccccc}
P & \xrightarrow{g'} & X & \xrightarrow{h'} & A \\
\downarrow f' & & \downarrow \lrcorner f=v' & & \downarrow v \\
Y & \xrightarrow{g} & Z & \xrightarrow{h} & B
\end{array}$$

That is, in the diagram on the right, assume that X is the pullback of A and Z over B . Then P is the pullback of the diagram on the left if and only if P is the pullback of the large outer diagram, namely if and only if

$$\begin{array}{ccc}
P & \xrightarrow{h' \circ g'} & A \\
\downarrow f' & & \downarrow v \\
Y & \xrightarrow{h \circ g} & B
\end{array}$$

is a pullback diagram.

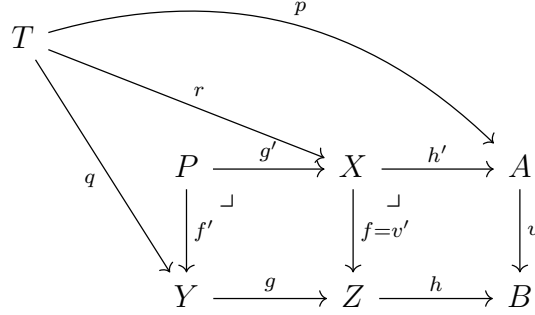
Proof. First suppose that P is the pullback of the small diagram. To prove the universal property of the large diagram, suppose that continuous maps $p: T \rightarrow A$ and $q: T \rightarrow Y$ from a topological space T make the diagram commute.

$$\begin{array}{c}
\begin{array}{c} T \\ \downarrow q \end{array} \begin{array}{c} \xrightarrow{p} \\ \searrow \end{array} \begin{array}{c} A \\ \downarrow v \end{array} \\
\begin{array}{ccccc}
P & \xrightarrow{g'} & X & \xrightarrow{h'} & A \\
\downarrow f' & \lrcorner & \downarrow \lrcorner f=v' & & \downarrow v \\
Y & \xrightarrow{g} & Z & \xrightarrow{h} & B
\end{array}
\end{array}$$

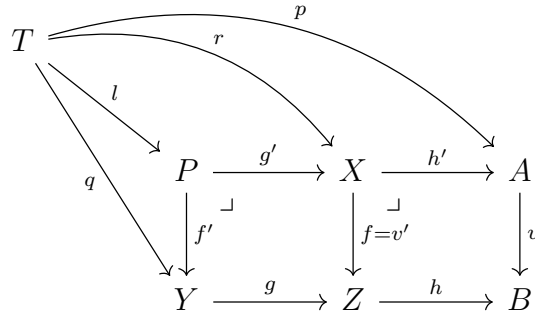
Then $p: T \rightarrow A$ and $g \circ q: T \rightarrow Z$ make the diagram consisting of $v: A \rightarrow B$ and $h: Z \rightarrow B$ commute.

$$\begin{array}{c}
\begin{array}{c} T \\ \downarrow q \end{array} \begin{array}{c} \xrightarrow{p} \\ \searrow \end{array} \begin{array}{c} A \\ \downarrow v \end{array} \\
\begin{array}{ccccc}
P & \xrightarrow{g'} & X & \xrightarrow{h'} & A \\
\downarrow f' & \lrcorner & \downarrow \lrcorner f=v' & & \downarrow v \\
Y & \xrightarrow{g} & Z & \xrightarrow{h} & B
\end{array}
\end{array}$$

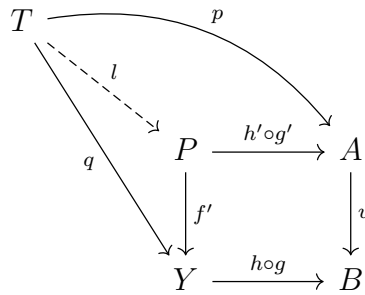
Since X is a pullback, there exists a unique map $r: T \rightarrow X$ making the diagram commute.



Applying the universal property of P to $r: T \rightarrow X$ and $q: T \rightarrow Y$, we obtain a unique map $l: T \rightarrow P$.



Since all of the above diagrams commute, the following diagram also commutes.



Uniqueness also follows from the uniqueness of r and l (why?). Thus P is $A \times_B Y$. We omit the converse. If one retraces the above operation in reverse, it should work. $\square$

By turning the diagram vertically, we also obtain the following.

Theorem 4. *Consider the situation of the following commutative diagram.*

$$\begin{array}{ccc}
P & \xrightarrow{g'} & X \\
\downarrow f' & & \downarrow f \\
Y & \xrightarrow{g=h'} & Z \\
\downarrow v' \lrcorner & & \downarrow v \\
A & \xrightarrow{h} & B
\end{array}$$

Then P is the pullback of the small square if and only if it is the pullback of the large outer diagram.

$$\begin{array}{ccc}
P & \xrightarrow{g'} & A \\
\downarrow v' \lrcorner f' & & \downarrow v \circ f \\
Y & \xrightarrow{h} & B
\end{array}$$

2. PERFECT MAPS

First, we explain that the base change of an embedding is again an embedding. In what follows, the word “an embedding” means “a topological embedding”.

Proposition 5. *Consider the following pullback diagram.*

$$\begin{array}{ccc}
P & \xrightarrow{g'} & X \\
\downarrow f' \lrcorner & & \downarrow f \\
Y & \xrightarrow{g} & Z
\end{array}$$

If f is an embedding, then f' is also an embedding. Furthermore, if f is a closed embedding, then f' is also a closed embedding.

Proof. Since X is embedded, we may regard it as a subset of Z . By (A3) of Proposition 2, P is $g^{-1}(X)$. That is, we have

$$\begin{array}{ccc}
g^{-1}(X) & \xrightarrow{g'} & X \\
\downarrow f' \lrcorner & & \downarrow f \\
Y & \xrightarrow{g} & Z
\end{array}$$

Here f' is the inclusion map into Y . In particular, it is an embedding. Also, when f is a closed embedding, X is a closed subset of Z , so $g^{-1}(X)$ is a closed subset of Y . Thus f' is a closed embedding. This completes the proof. $\square$

Proposition 6. *Let X, Y be topological spaces, and let $f: X \rightarrow Y$ be a continuous map. Then the following hold.*

- (T1) *X is Hausdorff if and only if the diagonal map $\Delta_X: X \rightarrow X \times X$ is a closed embedding.*

(T2) If Y is Hausdorff, then the map to the graph $\Gamma_f: X \rightarrow X \times Y; x \mapsto (x, f(x))$ is a closed embedding.

Proof. Proof of (T1): This follows immediately from the equivalence between the diagonal being closed and X being T_2 .

Proof of (T2): This follows from the following commutative diagram.

$$\begin{array}{ccccc} X & \xrightarrow{\Gamma_f} & X \times Y & \xrightarrow{pr} & X \\ \downarrow f & \lrcorner & \downarrow f \times 1_Y & & \downarrow f \\ Y & \xrightarrow{\Delta_X} & Y \times Y & \xrightarrow{pr} & Y \end{array}$$

Here note that, by (A4) of Proposition 2, the outer square is a pullback, and by (A2) of the same proposition, the diagram on the right is a pullback. Then by Theorem 3, the square on the left is a pullback. At this point, using Proposition 5 and the fact that Δ_X is a closed embedding, we see that its base change Γ_f is also a closed embedding. Done. $\square$

Definition 7. A continuous map $f: X \rightarrow Y$ is called a **perfect map** if f is a closed map and, for every $y \in Y$, $f^{-1}(y)$ is compact.

Proposition 8. The following hold.

- (K1) If X is compact and Y is Hausdorff, then every continuous map $f: X \rightarrow Y$ is a perfect map.
- (K2) A space K is compact if and only if the map $K \rightarrow \{*\}$ to a point is a perfect map.
- (K3) A space K is compact if and only if, for every topological space S , the projection $pr: S \times K \rightarrow S$ is a closed map.

Proof. (K1) and (K2) follow from compact Hausdorff-ness.

Using the tube lemma (see Lemma 16 in Appendix), one obtains the assertion in (K3) that the projection is closed. Next, to complete the proof of (K3), for a non-compact space X , we construct a space Y such that the projection $\pi: X \times Y \rightarrow Y$ is not a closed map.

Since X is non-compact, there exists an open cover $\mathcal{C}$ of X which has no finite subcover. For this $\mathcal{C}$, define the cardinal κ by

$$(2.1) \quad \kappa = \min\{\text{card}(\mathcal{U}) \mid \mathcal{U} \subset \mathcal{C} \text{ and } \mathcal{U} \text{ is a cover of } X\}.$$

Here $\text{card}(A)$ denotes the cardinality of the set A . By the choice of $\mathcal{C}$, the cardinal κ is infinite. By the minimality of κ , we can take a cover $\mathcal{U}$ such that $\kappa = \text{card}(\mathcal{U})$. Indexing this cover by κ , write $\mathcal{U} = \{U_\alpha\}_{\alpha < \kappa}$. Now put $Y = [0, \kappa] = \kappa + 1$, and define a family of open subsets $\{V_\alpha\}_{\alpha < \kappa}$ of Y by $V_\alpha = (\alpha, \rightarrow) = (\alpha, \kappa]$. We also define a subset O of $X \times Y$ by

$$O = \bigcup_{\alpha < \kappa} U_\alpha \times V_\alpha.$$

By definition, for each $\alpha < \kappa$, we have $\kappa \in V_\alpha$. Since $\{U_\alpha\}_{\alpha < \kappa}$ is a cover, for every $x \in X$ we have $(x, \kappa) \in O$. Moreover, O is clearly an open subset of $X \times Y$, so

$$(2.2) \quad F = (X \times Y) \setminus O$$

is a closed subset of $X \times Y$. We also see that for every $x \in X$, we have $(x, \kappa) \notin F$. Thus the image under the projection $\pi: X \times Y \rightarrow Y$ satisfies $\kappa \notin \pi(F)$. Now let us show that $\pi(F) = Y \setminus \{\kappa\}$. Let $\beta \in \kappa$ be arbitrary. Since $\text{card}(\{U_\alpha\}_{\alpha < \beta}) = \text{card}([0, \beta)) < \kappa$, by the minimality of κ , the family $\{U_\alpha\}_{\alpha < \beta}$ is not a cover of X . Thus we can take $x \in X \setminus \bigcup_{\alpha < \beta} U_\alpha$. Then

$$(2.3) \quad (x, \beta) \notin \bigcup_{\alpha < \beta} U_\alpha \times V_\alpha.$$

On the other hand, if $\beta \leq \alpha$, then $\beta \notin V_\alpha = (\alpha, \kappa]$, so

$$(2.4) \quad (x, \beta) \notin \bigcup_{\beta \leq \alpha < \kappa} U_\alpha \times V_\alpha.$$

Therefore $(x, \beta) \notin O$, and hence $(x, \beta) \in F$. This means that $\beta \in \pi(F)$. Consequently, we obtain $\pi(F) = Y \setminus \{\kappa\}$.

Since κ is an infinite cardinal, it is a limit ordinal. Thus, in $Y = [0, \kappa]$, the point κ is an accumulation point; in particular, $\{\kappa\}$ is not open. Therefore $\pi(F) = Y \setminus \{\kappa\}$ is not closed. Consequently, the projection $\pi: X \times Y \rightarrow Y$ is not a closed map. This finishes the proof of (K3). $\square$

Remark. *The statement (K3) is known as Kuratowski-Mrówka's theorem.*

We now describe properties of perfect maps by using pullbacks.

Theorem 9. *Let X and Y be topological spaces, and let $f: X \rightarrow Y$ be a continuous map. Then the following are equivalent.*

- (P1) *f is a perfect map.*
- (P2) *For every space Z , $f \times 1_Z: X \times Z \rightarrow Y \times Z$ is a perfect map.*
- (P3) *For every $g: Y' \rightarrow Y$, the base change $f': X' \rightarrow X$ is a perfect map.*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ \downarrow f'^\perp & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$

- (P4) *For every space Z , $f \times 1_Z: X \times Z \rightarrow Y \times Z$ is a closed map.*

$$\begin{array}{ccc} X' & \longrightarrow & X \\ \downarrow f' \lrcorner & & \downarrow f \\ Y' & \xrightarrow{g} & Y \end{array}$$
$$\begin{array}{ccccc}
 & & \text{(P5)} & & \\
 & \nearrow & & \searrow & \\
 \text{(P1)} & \Longrightarrow & \text{(P3)} & \Longrightarrow & \text{(P2)} \Longrightarrow \text{(P4)} \Longrightarrow \text{(P1)}
 \end{array}$$
$$\begin{array}{ccccc}
f^{-1}(q) & \longrightarrow & X' & \longrightarrow & X \\
\downarrow & \lrcorner & \downarrow f' & \lrcorner & \downarrow f \\
\{*\} & \xrightarrow{p} & Y' & \xrightarrow{g} & Y \\
& & \searrow q & &
\end{array}$$
$$(2.5) \quad X' = \{ (x, a) \in X \times Y' \mid f(x) = g(a) \}$$
$$(2.6) \quad (f^{-1}(g(p)) \times \{p\}) \cap A = \emptyset.$$

Indeed, if there were an intersection, it would contradict $p \notin f'(A)$. Since f is perfect, $f^{-1}(g(p))$ is compact, and hence $f^{-1}(g(p)) \times \{p\}$ is also compact. By the tube lemma (see Lemma 16 in Appendix), there exist open sets $V \subset X$ and $W \subset Y'$ such that $(V \times W) \cap A = \emptyset$ and $f^{-1}(g(p)) \times \{p\} \subset V \times W$. Now, since f is a closed map, if we put $O = Y \setminus f(X \setminus V)$, then this is an open subset of Y (this operation called the small image). Put $P = g^{-1}(O) \subset Y'$, and set $G = P \cap W$. We now prove that G is a neighborhood of p and does not meet $f'(A)$. First, let us show that P is a neighborhood of p . Since $f^{-1}(g(p)) \in V$, we

have $f^{-1}(g(p)) \cap X \setminus V = \emptyset$. Hence $g(p) \notin f(X \setminus V)$. Therefore $g(p) \in Y \setminus f(X \setminus V)$, so $p \in g^{-1}(Y \setminus f(X \setminus V)) = P$. Thus G is a neighborhood of p . Next we show that $G \cap f'(A) = \emptyset$. For $q \in G$, we have $q \in W$ and $g(q) \in Y \setminus f(X \setminus V)$. Thus $g(q) \notin f(X \setminus V)$, so $f^{-1}(g(q)) \cap (X \setminus V) = \emptyset$. That is, $f^{-1}(g(q)) \subset V$. Hence $f^{-1}(g(q)) \times \{q\} \subset V \times W$, and since $(V \times W) \cap A = \emptyset$, we have $q \notin f'(A)$. Thus G and $f'(A)$ do not meet. By the arbitrariness of p , the complement of $f'(A)$ is open, so $f'(A)$ is closed. In other words, f' is a closed map.

Proof of (P3) $\implies$ (P2): Consider the following diagram. By (A2) of Proposition 2, this is a pullback. Therefore, by (P3), $f \times 1_Z$ is a perfect map.

$$\begin{array}{ccc} X \times Z & \xrightarrow{pr} & X \\ \downarrow f \times 1_Z & & \downarrow f \\ Y \times Z & \xrightarrow{pr} & Y \end{array}$$

The implication (P2) $\implies$ (P4) is obvious.

The implication (P3) $\implies$ (P5) is obvious.

Proof of (P5) $\implies$ (P4): This is the same as the proof of (P3) $\implies$ (P2). Namely, consider the following diagram. By (A2) of Proposition 2, this is a pullback. Therefore, by (P5), $f \times 1_Z$ is a closed map.

$$\begin{array}{ccc} X \times Z & \xrightarrow{pr} & X \\ \downarrow f \times 1_Z & & \downarrow f \\ Y \times Z & \xrightarrow{pr} & Y \end{array}$$

Proof of (P4) $\implies$ (P1): Taking $Z = \{*\}$, we see that f is a closed map. Next we prove that $f^{-1}(p)$ is compact. Let S be an arbitrary topological space.

$$\begin{array}{ccc} f^{-1}(p) & \longrightarrow & X \\ \downarrow & \lrcorner & \downarrow f \\ \{*\} & \xrightarrow{p} & Y \end{array}$$

The above diagram is a pullback, so the following diagram is also a pullback (by (A5) of Proposition 2).

$$\begin{array}{ccc} f^{-1}(p) \times S & \xrightarrow{h} & X \times S \\ \downarrow v & \lrcorner & \downarrow f \times 1_S \\ S & \xrightarrow{(p, 1_S)} & Y \times S \end{array}$$

By Proposition 5 and the fact that $(p, 1_S)$ is a closed embedding, h is also a closed embedding. Moreover, by (P4), $f \times 1_S$ is also a closed map. We now show that v is a closed map. If A is a closed subset of the upper-left space, then $\{p\} \times v(A) = (f \times 1_S)(h(A))$. By what we

have accumulated so far, the right-hand side is closed. Thus $\{p\} \times v(A)$ is closed in the subspace $\{p\} \times S$. Hence $v(A)$ is a closed subset of S , and it follows that v is a closed map. Therefore, by the arbitrariness of S and by (K3) of Proposition 8, $f^{-1}(p)$ is compact. Thus f is a perfect map. This completes the proof. $\square$

Corollary 10. *Let $f: X \rightarrow Y$ be a perfect map. Then for every subset B of Y , $f|_{f^{-1}(B)}: f^{-1}(B) \rightarrow B$ is a perfect map.*

Proof. Consider the following diagram. Here $v = f|_{f^{-1}(B)}$.

$$\begin{array}{ccc} f^{-1}(B) & \xrightarrow{h'} & X \\ \downarrow v & \lrcorner & \downarrow f \\ B & \xrightarrow{h} & Y \end{array}$$

Since this diagram is a pullback, v is a perfect map. $\square$

Lemma 11. *Let $f: X \rightarrow Y$ and $g: Y \rightarrow Z$ be perfect maps. Then $g \circ f: X \rightarrow Z$ is also perfect.*

Proof. Take an arbitrary $h: A \rightarrow Z$ and consider the following diagram.

$$\begin{array}{ccc} P & \xrightarrow{u'} & X \\ \downarrow f' \lrcorner & & \downarrow f \\ Q & \xrightarrow{u} & Y \\ \downarrow g' \lrcorner & & \downarrow g \\ A & \xrightarrow{h} & Z \end{array}$$

Then, by (P5) of Theorem 9, f' and g' are closed maps. Hence their composite $g' \circ f'$ is also a closed map.

Therefore, in the pullback diagram

$$\begin{array}{ccc} P & \xrightarrow{u'} & X \\ \downarrow g' \circ f' \lrcorner & & \downarrow g \circ f \\ Q & \xrightarrow{u} & Z \end{array}$$

the map $g' \circ f'$ is always closed, and so, again by 9, $g \circ f$ is perfect. $\square$

Corollary 12. *Let $f: X \rightarrow Y$ be a perfect map, and let K be a compact subset of Y . Then $f^{-1}(K)$ is a compact subset of X .*

Proof. Put $g = f|_{f^{-1}(K)}: f^{-1}(K) \rightarrow K$. This is a perfect map by Corollary 10. Also $K \rightarrow \{*\}$ is a perfect map by (K2) of Proposition 8. By Lemma 11, the composition of these two maps, $f^{-1}(K) \rightarrow \{*\}$, is perfect. Therefore, again by (K2) of Proposition 8, $f^{-1}(K)$ is compact. $\square$

Theorem 13 ([7, Theorem 1.1]). *Let X, Y be topological spaces, and let Z be a Hausdorff space. Let $f: X \rightarrow Y$ be a perfect map and let $g: X \rightarrow Z$ be a continuous map. Then $(f, g): X \rightarrow Y \times Z$ is a perfect map.*

Proof. The map $(1_X, g): X \rightarrow X \times Z$ is a closed embedding because Z is Hausdorff (Proposition 6). In particular, it is perfect. Moreover, $f \times 1_Z: X \times Z \rightarrow Y \times Z$ is a perfect map ((P2) of Theorem 9). Therefore their composite, (f, g) , is also a perfect map (Lemma 11). $\square$

Theorem 14 ([7, Corollary 1.4]). *Let A, B, C be topological spaces, and assume that B is Hausdorff. Let $\alpha: A \rightarrow B$ be a continuous map and let $\beta: B \rightarrow C$ be a perfect map. If $\beta \circ \alpha: A \rightarrow C$ is a perfect map, then α is also a perfect map.*

Proof. I could not think of a proof using pullbacks, so I give an ordinary proof. Put $\gamma = (\alpha, \beta \circ \alpha): A \rightarrow B \times C$. Since B is Hausdorff, γ is a perfect map (Theorem 13). Let $G_\beta \subset B \times C$ be the graph of β . Then $pr: G_\beta \rightarrow B$ is a homeomorphism. In particular, it is perfect. Since $\gamma(A) \subset G_\beta$ and $\alpha = pr \circ \gamma$, it follows that α is perfect. $\square$

In this note, when we say a space is completely regular, we mean it in the sense that assumes T_1 . Also, βX denotes the Stone–Cech compactification of a topological space X .

Theorem 15. *Let X and Y be completely regular spaces, and let $f: X \rightarrow Y$ be a continuous map. Then the following are equivalent.*

- (Q1) *f is a perfect map.*
- (Q2) *There exist a compact space K and a closed embedding $i: X \rightarrow Y \times K$ such that $f = pr \circ i$. Here $pr: Y \times K \rightarrow Y$.*
- (Q3) *The following diagram is a pullback. Here g and g' are the natural embeddings.*

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow g' & \lrcorner & \downarrow g \\ \beta X & \xrightarrow{\beta f} & \beta Y \end{array}$$

- (Q4) *Let an arbitrary compactification of Y be denoted by ηY . Then the following diagram is a pullback.*

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow g' & \lrcorner & \downarrow g \\ \beta X & \xrightarrow{\beta f} & \eta Y \end{array}$$

- (Q5) *Let γX and ηY be compactifications of X and Y , respectively, which moreover make the following diagram commute. Then*

the following diagram is a pullback.

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow g' \lrcorner & & \downarrow g \\ \gamma X & \xrightarrow{\gamma f} & \eta Y \end{array}$$

(Q6) In the same situation as condition (Q3), $(\beta f)^{-1}(Y) = X$ holds.

(Q7) In the same situation as condition (Q4), $(\beta f)^{-1}(Y) = X$ holds.

(Q8) In the same situation as condition (Q5), $(\gamma f)^{-1}(Y) = X$ holds.

Proof. Proof of (Q1) $\implies$ (Q2): Consider the following commutative diagram.

$$\begin{array}{ccccc} X & \xleftarrow{\Delta} & X \times \beta X & \xrightarrow{pr} & X \\ & \searrow f' \circ \Delta & \downarrow f' = \bar{f} \times 1_{\beta X} & & \downarrow f \\ & & Y \times \beta X & \xrightarrow{pr} & Y \end{array}$$

The morphism extending from X to X becomes the identity map when composed. Thus, taking $K = \beta X$, (Q2) is satisfied.

Proof of (Q2) $\implies$ (Q1): This follows from (K3) of Proposition 8 and Lemma 11.

Proof of (Q3) $\vee$ (Q4) $\vee$ (Q5) $\implies$ (Q1): Maps such as βf and γf are continuous maps between compact Hausdorff spaces, hence are perfect maps. The conclusion follows from this and (P3) of Theorem 9.

Proof of (Q1) $\implies$ (Q3) $\wedge$ (Q4) $\wedge$ (Q5):

Since (Q5) $\implies$ (Q3) $\wedge$ (Q4), it is enough to prove (Q1) $\implies$ (Q5).

Consider the following diagram. After constructing Q , we construct P .

$$\begin{array}{ccc} P & \xrightarrow{h'} & X \\ \downarrow f' \lrcorner & & \downarrow f \\ Q & \xrightarrow{h} & Y \\ \downarrow u' \lrcorner & & \downarrow g \\ \gamma X & \xrightarrow{\gamma f} & \eta Y \end{array}$$

Since $g: Y \rightarrow \eta Y$ is an embedding, Q is homeomorphic to $(\gamma f)^{-1}(Y)$. Thus we obtain the following diagram.

$$\begin{array}{ccc} P & \xrightarrow{h'} & X \\ \downarrow f' \lrcorner & & \downarrow f \\ (\gamma f)^{-1}(Y) & \xrightarrow{h} & Y \\ \downarrow u' \lrcorner & & \downarrow g \\ \gamma X & \xrightarrow{\gamma f} & \beta Y \end{array}$$

Here u' is the inclusion map.

By (P3) of the preceding Theorem 9, f' is perfect. Similarly, γf is perfect, so h is perfect, and h' is also perfect.

Since there is the obvious embedding $v': X \rightarrow \beta X$, the following diagram holds.

$$\begin{array}{ccc}
 X & \xrightarrow{1_X} & X \\
 \downarrow v' & & \downarrow f \\
 & P \xrightarrow{h'} X & \\
 & \downarrow f' \lrcorner & \\
 & (\gamma f)^{-1}(Y) \xrightarrow{h} Y & \\
 & \downarrow u' \lrcorner & \downarrow g \\
 & \gamma X \xrightarrow{\gamma f} \eta Y &
 \end{array}$$

Therefore, by the universal property of Q , there exists a unique $r: X \rightarrow Q$.

$$\begin{array}{ccc}
 X & \xrightarrow{1_X} & X \\
 \downarrow v' & & \downarrow f \\
 & P \xrightarrow{h'} X & \\
 & \downarrow f' \lrcorner & \\
 & (\gamma f)^{-1}(Y) \xrightarrow{h} Y & \\
 & \downarrow u' \lrcorner & \downarrow v \\
 & \gamma X \xrightarrow{\gamma f} \eta Y &
 \end{array}$$

Furthermore, by the universal property of P , there exists a unique $l: X \rightarrow P$.

$$\begin{array}{ccc}
 X & \xrightarrow{1_X} & X \\
 \downarrow v' & & \downarrow f \\
 & P \xrightarrow{h'} X & \\
 & \downarrow f' \lrcorner & \\
 & (\gamma f)^{-1}(Y) \xrightarrow{h} Y & \\
 & \downarrow u' \lrcorner & \downarrow v \\
 & \gamma X \xrightarrow{\gamma f} \eta Y &
 \end{array}$$

Here $g' \circ l = 1_X$, and g' and 1_X are perfect maps. Therefore, by Theorem 14, l is also a perfect map. Furthermore, since $r = f' \circ l$, r is also a perfect map.

Now $X \subset (\gamma f)^{-1}(Y)$, so there is an inclusion map $X \rightarrow (\beta f)^{-1}(Y)$. If we replace r by this map in the diagram, the commutative diagram is still satisfied. Thus, by the uniqueness of r , the map r is the inclusion map $X \rightarrow (\gamma f)^{-1}(Y)$. Since X is dense in γX , it is in particular dense in $(\gamma f)^{-1}(Y)$. We have now seen that r is a perfect map, hence in particular a closed map, so X is a closed subset of $(\gamma f)^{-1}(Y)$. By density, $X = (\gamma f)^{-1}(Y)$, and in the end r was the identity map. Moreover, by commutativity of the diagram, $h = f$. Thus we finally obtain the following diagram.

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow & \lrcorner & \downarrow \\ \gamma X & \xrightarrow{\gamma f} & \eta Y \end{array}$$

This means that (Q5) holds.

Proof of (Q5) $\implies$ (Q6) $\wedge$ (Q7) $\wedge$ (Q8): Since (Q8) $\implies$ (Q7) $\wedge$ (Q8) holds, it is enough to prove (Q5) $\implies$ (Q8). This is similar to the proof above of (Q1) $\implies$ (Q5), but let us explain it anyway instead of being lazy.

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow & \lrcorner & \downarrow \\ \gamma X & \xrightarrow{\gamma f} & \eta Y \end{array}$$

is a pullback, and

$$\begin{array}{ccc} (\gamma f)^{-1}(Y) & \xrightarrow{(\gamma f|_{f^{-1}(Y)})} & Y \\ \downarrow & \lrcorner & \downarrow \\ \gamma X & \xrightarrow{\gamma f} & \eta Y \end{array}$$

is also a pullback. It follows that $X = (\gamma f)^{-1}(Y)$.

Proof of (Q6) $\vee$ (Q7) $\vee$ (Q8) $\implies$ (Q1): This follows from the fact that βf and γf are perfect maps and from Corollary 10.

This completes the proof. $\square$

Remark. The conditions (Q6), (Q7), and (Q8) of Theorem 15 are respectively equivalent to

- (1) $\beta f(\beta X \setminus X) = \text{cl}_{\beta Y}(f(X)) \setminus f(X)$,
- (2) $\beta f(\beta X \setminus X) = \text{cl}_{\eta Y}(f(X)) \setminus f(X)$,
- (3) $\gamma f(\gamma X \setminus X) = \text{cl}_{\eta Y}(f(X)) \setminus f(X)$.

When f is surjective, these become

- (1) $\beta f(\beta X \setminus X) = \beta Y \setminus Y$,

- (2) $\beta f(\beta X \setminus X) = \eta Y \setminus Y,$
(3) $\gamma f(\gamma X \setminus X) = \eta Y \setminus Y.$

In other words, the image of the remainder of the compactification is equal to the remainder of the target.

3. APPENDIX

Lemma 16 (The tube lemma). *Let X be a topological space, K be a compact space, $x \in X$ and U be an open set of $X \times K$. If U contains $\{x\} \times K$, then there exists a neighborhood N of x such that $N \times K \subset U$.*

Proof. Since K is compact and U is open, we can take finite open neighborhoods $O_1, \dots, O_n$ of x and open subsets $V_1, \dots, V_n$ such that

$$(3.1) \quad \{x\} \times K \subset \bigcup_{i=1}^n O_i \times V_i \subset U.$$

Putting $N = \bigcap_{i=1}^n O_i$ finishes the proof. $\square$

A continuous map f is called a **proper map** if the inverse image under f of every compact set is compact.

A space X is called a **k -space** if, for a subset A of X , A is closed if and only if, for every compact subset K , $K \cap A$ is closed in K . In other words, the quotient topology induced by the family of maps $\{\iota_K: K \rightarrow X\}_{K: \text{cpt in } X}$ agrees with the topology of X itself. Such a space is also called **compactly generated**.

A topological space X is called **weakly Hausdorff** if every compact subset is closed.

Proposition 17. *Let X be a topological space, and let Y be a compactly generated weakly Hausdorff space. Then a continuous map $f: X \rightarrow Y$ is a perfect map if and only if it is a proper map.*

Proof. First assume that f is perfect. Then, by Corollary 12, f is a proper map. Conversely, assume that f is proper. Then, by the definition of a proper map, each fiber of f is compact. We now show that f is a closed map. Let A be an arbitrary closed subset of X . It is enough to show that $f(A)$ is closed in Y . Let K be an arbitrary compact subset of Y . Then

$$(3.2) \quad f(A) \cap K = f(A \cap f^{-1}(K))$$

holds. (This formula is called the projection formula, and it always holds for a map between sets and their subsets.) Since f is proper, $f^{-1}(K)$ is compact, and since A is closed in X , $A \cap f^{-1}(K)$ is closed in $f^{-1}(K)$. In particular, it is compact. Therefore $f(A \cap f^{-1}(K))$ is also compact. Thus $f(A) \cap K$ is a compact subset of Y . Since Y is weak Hausdorff, $f(A) \cap K$ is closed. By the arbitrariness of K and the fact that Y is a k -space, $f(A)$ is closed. Hence f is a closed map, and consequently a perfect map. $\square$

Remark. In [1], Tomoki Yuji explained to me the properties of perfect maps with pullbacks, and every space was assumed to be completely regular. Perhaps the category of completely regular spaces has a somewhat scheme-like flavor. In the present document, I have modified the argument so that, except for the final theorem, the proofs are carried out as much as possible for general topological spaces. As a result, the proof of Theorem 9 may have become a strange one in which the implications are proved one by one.

Remark. According to [7], (Q2) of Theorem 15 seems to have been found in [6, Theorem 1] and [8, Theorem, p.21]. According to [5], (Q3) of Theorem 15 seems to have been found in [4, Lemma 1.5].

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